

David Neish



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SUMMARY:

Solutions Engineer with 5+ years building distributed systems in finance, focused on applied AI and accelerated infrastructure. I build software systems that people use, and businesses profit from. Obtaining an M.S. in Human-Computer Interaction in Spring 2027 and researching human-centered agent design and multimodal feedback architectures.

SKILLS

Languages & Tools: Java, Maven, Python, Pytorch, SQL, Git, gRPC, REST, NATS, Micrometer

AI/ML Frameworks: Langchain, LangGraph, LlamaIndex, trl, NVIDIA NIM, Nemo

Cloud & DevOps: AWS, EKS, GCP, Google Cloud Run, Azure AI Foundry, Docker, Kubernetes, GitHub Actions

Data & Databases: SQL BigQuery, Oracle DB, NoSQL, Hazelcast, Ignite

Finance-Specific: ISO 8583, ISO 20022, Protobuf

EXPERIENCE

Applied AI (Software Engineer 2) | MASTERCARD | AUGUST 2021 – PRESENT

- Designed and maintained distributed, event-driven applications using NATS for real-time pub/sub messaging, supporting mission-critical financial workflows processing 10M+ daily transactions and contributing to the system reliability needed for Mastercard's transition toward real-time clearing and faster settlement cycles.
- Architected and optimized a Databricks + LangChain retrieval pipeline (BM25, metadata tagging, Llama-70B), improving task success rate from 0% to 92% after redesigning retrieval and embedding architecture. Enabled 20+ global sales specialists to access case studies faster, accelerating campaign delivery by 60% and improving credit-card activation targeting.
- Architected an AI-powered Python web app using NiceGui and GPT-4 to reduce language translation time for PowerPoint, Docs, and PDFs by 95%, with an error rate under 5%. Realized cost savings of up to 99% compared to traditional agency-based translations.
- Developed a containerized testing framework using Testcontainers to reproduce distributed-system behaviors such as network partitions and split-brain cache states. Enabled earlier fault detection and improved confidence across teams responsible for high-volume event-driven services.
- Served as an AI adoption advisor, briefing executives on coding agents, LLM strategy, and industry direction. Built and manage a 37-model governance registry that standardizes evaluation, mitigates risk, and accelerates enterprise AI integration.
- Built an automated CI/CD pipeline for registering Protobuf schemas, replacing a manual PR process with auto-approvals via Bitbucket's API. Integrated Salesforce Maven Plugin to verify backward compatibility, saving 250+ hours quarterly and enforcing consistent, production-safe schema updates.

SOFTWARE ENGINEER INTERN | FIDELITY INVESTMENTS | JUNE 2020 – AUGUST 2020

- Documented cloud adoption patterns and developed an AWS S3 onboarding guide, leveraging CLI and SDK workflows to simplify file transfers and storage setup for new teams.

- Built a Java Spring Boot CLI utility that automated common S3 bucket operations, reducing manual steps and easing developer onboarding.

Education

SUNY Fredonia | B.S. Computer Science | May 2021

Indiana University | M.S. Human-Computer Interaction | May 2027 |

Certifications

AWS Certified Solutions Architect – Associate | Azure AI Fundamentals | NVIDIA Generative AI with Diffusion Models | NVIDIA AI & Infrastructure and Operations Fundamentals | NVIDIA Fundamentals of Deep Learning

Applied AI Projects & Agentic Systems

finplatform — Built and deployed a production fintech platform on **Google Cloud Run (Python, React/Vite)** that consolidates equity research workflows. Implemented a tiered LLM router with budget-aware fallback cascades (Claude Sonnet → Claude Haiku → local Ollama Qwen 2.5 → deterministic code) governed by user credit tiers. Built an automated research pipeline with SSRF-safe web retrieval, deduplication, and a strict 3-source corroboration requirement before extracting structured data.

research-digest-mcp — Built an opensource Model Context Protocol (**MCP**) server, capable of indexing 20,000+ papers, exposing tools for LLM clients to search, track topic trends, and fetch digests. Implemented a zero-dependency Python standard library core with optional NumPy/Scikit-learn embeddings. Built arXiv ingestion to handle rate limits, using publication-date validation to prevent missing data.

Passage — Built a document and voice translation platform on Google Cloud Run that preserves PDF layout structure alongside **live text and audio translation**. Built a provider-agnostic router over OpenAI-compatible endpoints to swap between hosted APIs and a local open-weight Gemma model on a local GPU based on cost and latency needs. Replaced fragile prompt-based rules with deterministic masking code to guarantee names, numbers, and formatting remained unaltered during translation. Evaluated model variants across 500+ CI tests, finding task-specialized small models beat larger reasoning models in speed and translation accuracy.